

QVerse — Week 11: Grover Search and Amplitude Amplification

Theory Booklet | Introduction to Quantum Computing & Programming

QVerse

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How this booklet fits the course

This theory booklet is designed to be read alongside `11_grover_amplitude_amplification.ipynb`. It contains the conceptual and mathematical material that should be explained during Week 11; the notebook then turns those ideas into experiments, circuit construction, and exercises.

Learning objectives

- Explain a marked state and phase oracle.
- Implement a two-qubit Grover search.
- Build a diffuser.
- State Grover’s quadratic, not exponential, query speedup.

Recommended workflow

1. Read the relevant theory section.
2. Predict the result of the corresponding notebook experiment.
3. Run the notebook.
4. Explain any difference between the prediction and the output.
5. Complete the core exercises without treating the solution notebook as a substitute for reasoning.

1. The search problem

Suppose there are

$$N = 2^n$$

possible basis states and M of them are marked as solutions.

An unstructured classical search may require $O(N/M)$ oracle queries in the worst-case scaling.

Grover’s algorithm finds a marked item using approximately

$$O\left(\sqrt{\frac{N}{M}}\right)$$

oracle queries.

For one marked state,

$$O(\sqrt{N}).$$

This is a **quadratic** speedup, not an exponential one.

2. Uniform superposition

Starting from

$$|0\rangle^{\otimes n},$$

apply Hadamards:

$$|s\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$

Every basis state initially has equal amplitude

$$\frac{1}{\sqrt{N}}$$

and equal probability

$$\frac{1}{N}.$$

The algorithm must selectively increase the amplitude of marked states.

3. Phase oracle

The Grover oracle marks a solution by changing its phase.

For one target state $|w\rangle$,

$$O_w|w\rangle = -|w\rangle,$$

while for unmarked states,

$$O_w|x\rangle = |x\rangle.$$

The oracle therefore changes **phase**, not probability.

Immediately after the oracle, a Z-basis histogram is unchanged. The marked state still has the same squared amplitude.

A second operation is required to convert the phase difference into amplitude amplification.

4. The diffuser

The Grover diffuser is

$$D = 2|s\rangle\langle s| - I.$$

Geometrically, it is a reflection about the uniform-superposition state $|s\rangle$.

In amplitude language, it acts like an **inversion about the mean**.

The oracle first reflects the state with respect to the unmarked subspace. The diffuser performs a second reflection. Two reflections produce a rotation toward the marked-state direction.

5. One two-qubit example

For $n = 2$,

$$N = 4.$$

The uniform state is

$$|s\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

Suppose $|11\rangle$ is marked. After the oracle:

$$\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

The mean amplitude is

$$\bar{a} = \frac{1/2 + 1/2 + 1/2 - 1/2}{4} = \frac{1}{4}.$$

Inversion about the mean maps each amplitude a approximately as

$$a \rightarrow 2\bar{a} - a.$$

Then the three unmarked amplitudes become

$$2(1/4) - 1/2 = 0,$$

and the marked amplitude becomes

$$2(1/4) - (-1/2) = 1.$$

So one Grover iteration finds the unique target with probability 1 in the ideal two-qubit case.

6. Geometric two-dimensional picture

Even though the Hilbert space has dimension N , Grover dynamics remain in a two-dimensional subspace spanned by:

- a normalized superposition of marked states $|w\rangle$;
- a normalized superposition of unmarked states $|r\rangle$.

Write

$$|s\rangle = \sin \theta |w\rangle + \cos \theta |r\rangle,$$

with

$$\sin^2 \theta = \frac{M}{N}.$$

One Grover iteration rotates the state by approximately

$$2\theta$$

toward the marked direction.

This geometric picture explains both the speedup and the possibility of applying too many iterations.

7. Optimal number of iterations

For $M \ll N$, the optimal number is approximately

$$k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{M}}.$$

The success probability oscillates. Grover amplification is not monotonic forever.

If the circuit continues rotating beyond the marked direction, the success probability decreases again.

This explains why “more Grover iterations” is not automatically better.

8. Multiple marked states

When $M > 1$, the same amplitude-amplification framework applies to the whole marked subspace.

The iteration count scales as

$$O\left(\sqrt{\frac{N}{M}}\right).$$

If M is unknown, choosing the optimal number of iterations becomes more complicated.

For the course, the notebook focuses on one marked item so the underlying mechanism remains clear.

9. Building a simple phase oracle

For a small number of qubits, one can mark a computational-basis target by:

1. applying X gates to map the desired target to an all-ones pattern;
2. applying an appropriate multi-controlled phase operation;
3. undoing the X gates.

The precise construction depends on gate conventions and qubit ordering.

Because Week 5 established Qiskit bit ordering, students should verify each oracle against the intended target instead of trusting visual intuition.

10. Why removing the diffuser fails

If we apply only the phase oracle, the amplitudes differ by signs but their magnitudes are unchanged.

Therefore computational-basis measurement still sees a uniform probability distribution.

The diffuser converts hidden phase information into a changed amplitude distribution.

This is a direct continuation of Week 4: phase matters because a later interference step can make it observable.

11. General amplitude amplification

Grover search is a special case of **amplitude amplification**.

If a unitary A prepares a state containing a “good” component and an oracle can reflect the good subspace, repeated reflections can amplify the good amplitude.

This perspective makes Grover useful beyond literal database search.

12. Complexity claim

The clean statement is:

Grover provides a quadratic query speedup for unstructured search.

Avoid saying:

- “Grover is exponentially faster”;
- “Grover searches all states at once and reads the right one”;
- “Grover makes NP-complete problems easy.”

Quadratic speedup can still be valuable, but it does not remove exponential scaling for many combinatorial problems.

13. Common misconceptions

“The oracle makes the target more probable”

The oracle only changes phase.

“The diffuser knows which answer is correct”

It is target-independent. The oracle contains the problem-dependent marking.

“More iterations always help”

Success probability oscillates.

“Grover gives exponential speedup”

It gives quadratic query speedup.

14. What to know before the notebook

You should be able to explain:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_x |x\rangle,$$

$$O_w|w\rangle = -|w\rangle,$$

$$D = 2|s\rangle\langle s| - I,$$

and

$$k_{\text{opt}} \approx \frac{\pi}{4} \sqrt{\frac{N}{M}}.$$

Most importantly, you should understand the sequence:

mark phase $\rightarrow$ interfere $\rightarrow$ amplify probability.

Suggested references

- IBM Quantum Learning, *Basics of Quantum Information* and current Qiskit learning material: <https://quantum.cloud.ibm.com/learning/en/courses>
- IBM Quantum Documentation and Qiskit guides: <https://quantum.cloud.ibm.com/docs>
- Quantum Rings, *Quantum 101*: <https://www.quantumrings.com/course/quantum101>
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.

The software-facing details in the practical notebook should be treated as the canonical implementation for this course. The purpose of this booklet is to explain the physics, mathematics, and algorithmic reasoning behind that implementation.